Managing I/O Operations

Subject: Programming for Problem Solving

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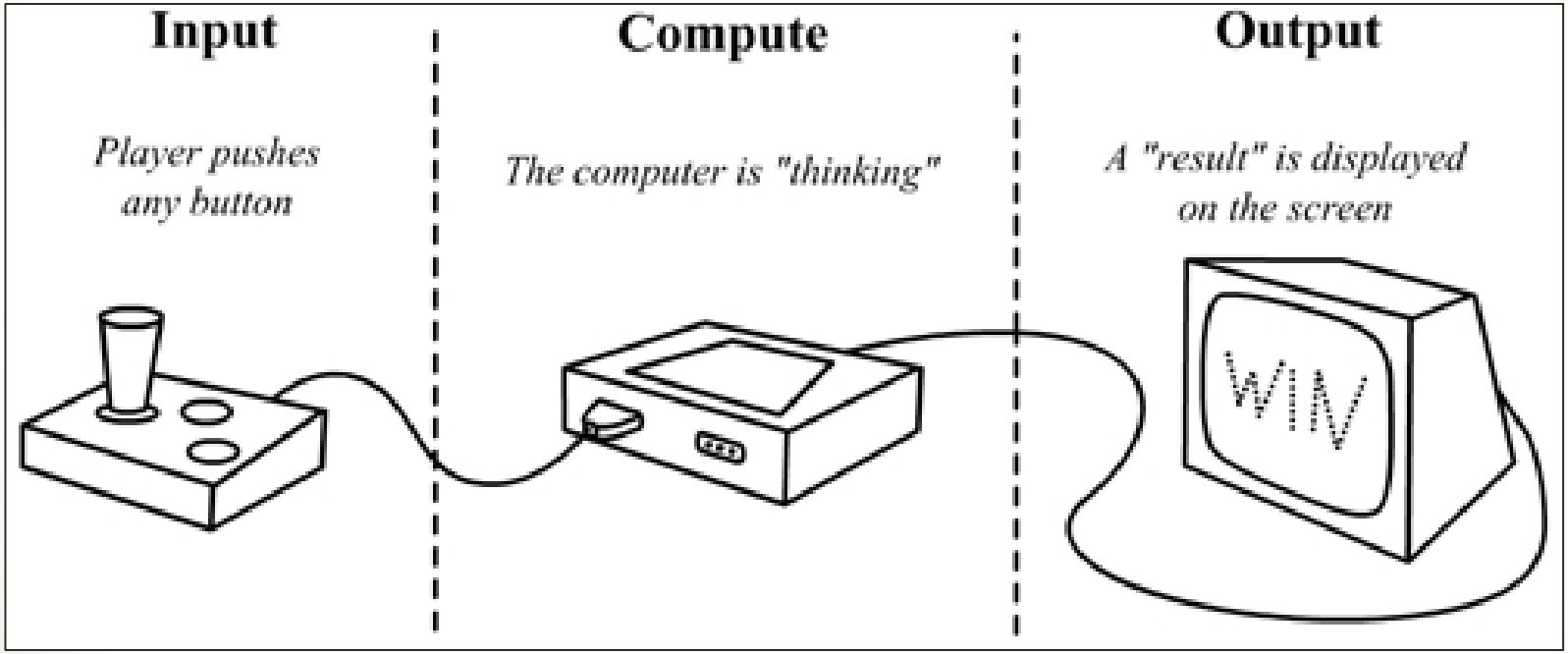
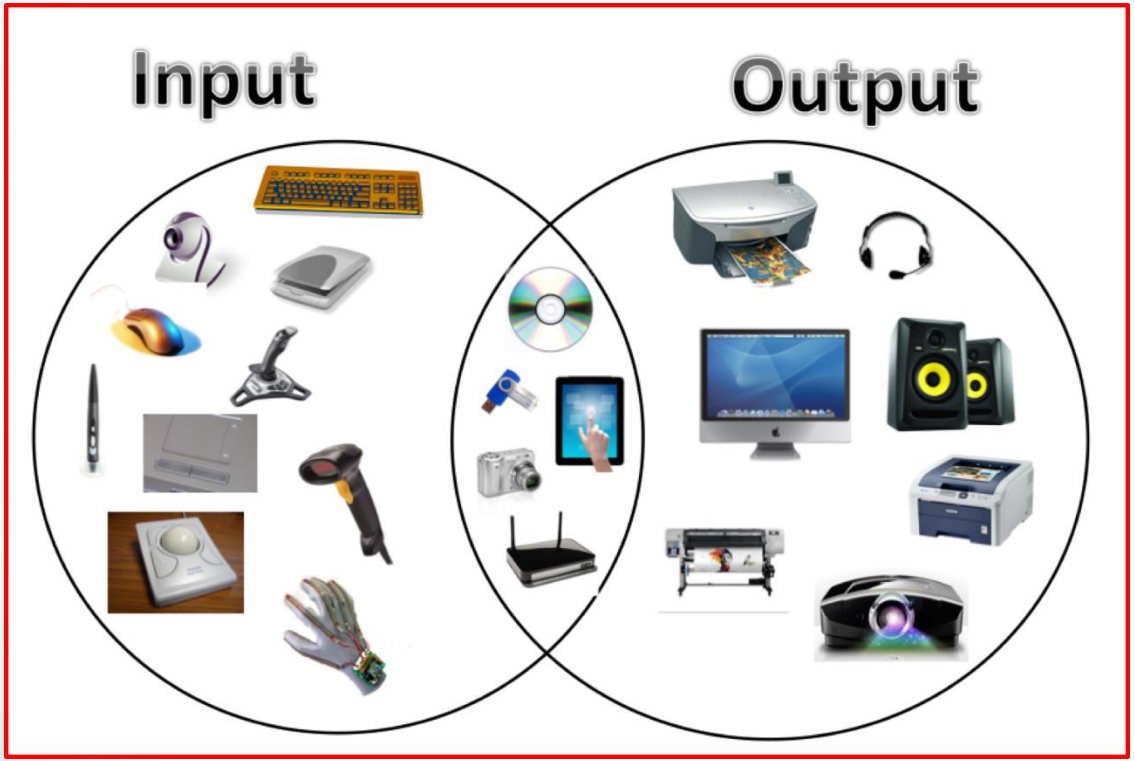
Introduction

- Input / output (I/O) is an essential part of any computer program for interaction of user and machine.
- Most of the higher level languages support built-in I/O functions.
- C has I/O functions as part of its library which mainly include "**stdio.h**" and "**conio.h**" header files.

I/O Operation



Supporting Devices



Reading a Character

- To read a single character input from user ‘getchar()’ function is used.
- Syntax:

  var_name = getchar();

- E.g:

```
char user_input;
user_input=getchar();
```

Character Test Functions

Function	Description	Example
isalnum()	This function is used to check whether a character is an alphanumeric(i.e A to Z or a to z) or (0 to 9).	isalnum('a')=True(True means non zero value). isalnum('&')=False
isalpha()	This function is used to test whether the given character is alphabetic character or not.	isalpha('a')=True(1) isalpha('5')=False(0)
isdigit()	This function is used to check whether or not it is a digit(0 to 9).	isdigit('9')=True isdigit('a')=False
islower()	This function is used to check whether the given character is lower case or not(a to z).	islower('a')=True islower('A')=False
isupper()	This function is used to check whether the given character is uppercase letter or not(A to Z).	isupper('a')=False Isupper('A')=True

Writing a Character

- To write one character at a time to the terminal "**putchar()**" function is used in C.
- Syntax:

  putchar(var_name);

- E.g:

```
char answer='Y';
putchar(answer);


putchar('\n');
```

Formatted Input

- **scanf()** function is used to take the user input in various ways.
- Syntax:

  scanf (“control string”, arg1, arg2, …, argn);

- Control string specifies the format in which data has to be entered.
- %d ← Signed / Unsigned int
- %ld ← Long int
- %hd ← Short int

Inputting Integer Numbers

- Syntax:

  scanf(“%wd1 %wd2 %wd3 …”, &var1, &var2, &var3,…);

- E.g:

scanf(“%2d %5d %d”,&num1, &num2, &num3);

- Case1: 50 32651 888
- Case2: 32651 50 888

scanf(“%d %*d %d”,&data1,data2);

- Case3: 123 456 789

Case3

datat1
123
data2
789

Case1	Case2
num1	num1
50	32
num2	num2
32651	651
num3	num3
888	50

Inputting Real Numbers

- Unlike int numbers, width of real numbers can not be specified.
- Float ← %f
- double ← %lf
- E.g:
- **scanf (“%f %f %f”, &x, &y, &z);**
- Case1: 123.45 4425.98E-2 66
- You can skip reading a real number using %*f specification.

Case1

x
123.45
y
44.2598
z
66.0

Inputting Character Strings

- For single character ← **getchar();**
- For multiple characters ← **scanf("%s",var);**
- **%wc** or **%ws** is used to specify number of characters while scanning string.
- Some other specifications of **scanf** function is as following:
- **%[characters]** ← Only allows specified characters
- **%[^characters]** ← Exactly reverse of above

Reading Blank Spaces

- Specifier %s can not be used with strings having blank spaces.
- But one can include white spaces in scanning using %[] specification.

```
main()
{
   char address[80];
   printf("Enter address\n");
   scanf("%[^\n]", address);
   printf("%-80s", address);
}
```

Output

```
Enter address
New Delhi 110 002
New Delhi 110 002
```

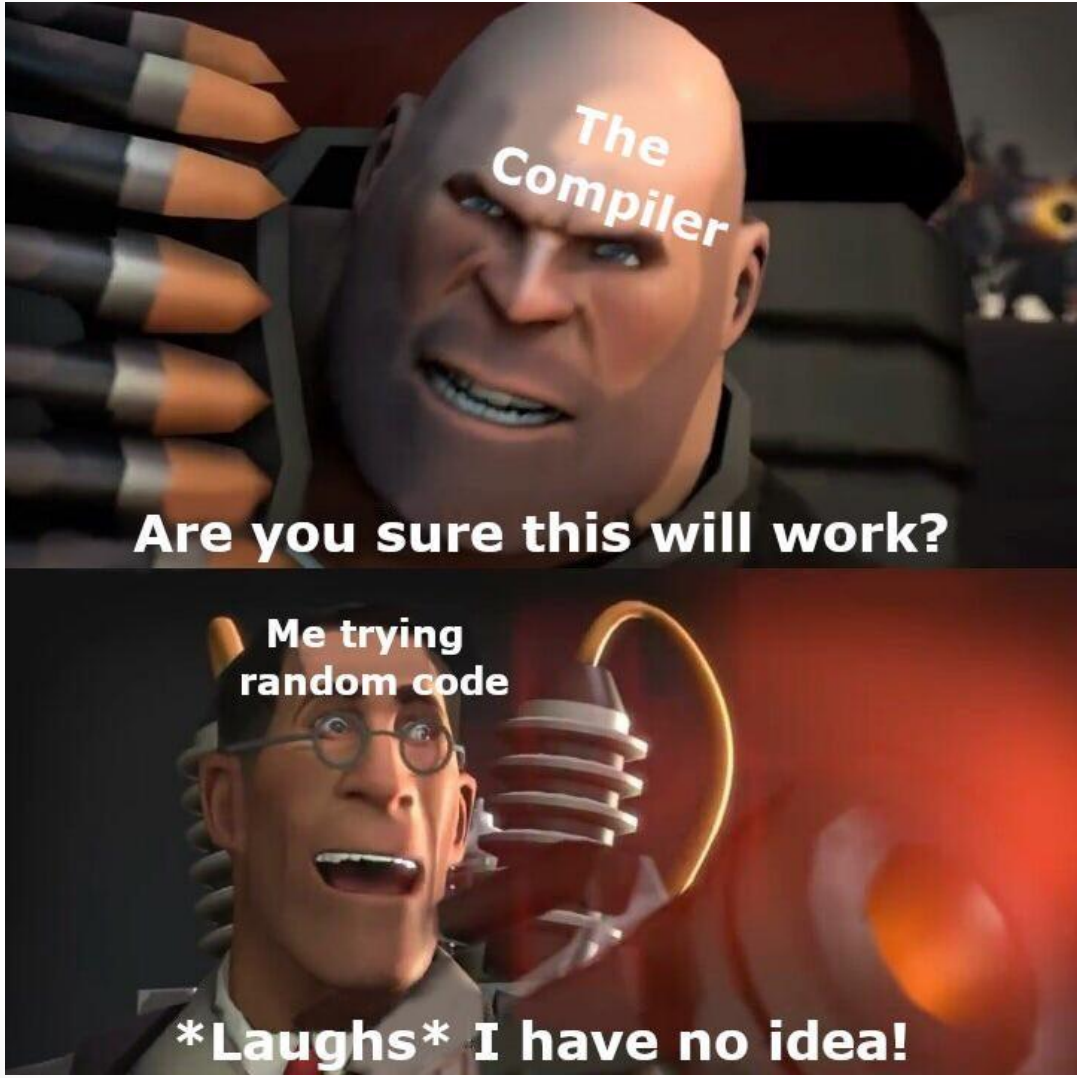
Error Detection in Input

- When a **scanf()** function completes reading list, it returns the value for every successful read.

  E.g.: scanf("%d %f %s", &a,&b,name);

- Above statement will return value 3 with following value assignment.

  20 150.25 motor


The Compiler
Are you sure this will work?
Me trying random code
Laughs I have no idea!

Error Detection in Input

```
main()
{
   int a;
   float b;
   char c;
   printf("Enter values of a, b and c\n");
   if (scanf("%d %f %c", &a, &b, &c) == 3)
      printf("a = %d b = %f c = %c\n" , a, b, c);
   else
      printf("Error in input.\n");
}
```

```
Enter values of a, b and c
    12 3.45 A
    a = 12    b = 3.450000     c = A
    Enter values of a, b and c
    23 78 9
    a = 23    b = 78.000000    c = 9
    Enter values of a, b and c
    8 A 5.25
    Error in input.
    Enter values of a, b and c
    Y 12 67
    Error in input.
    Enter values of a, b and c
    15.75 23 X
    a = 15    b = 0.750000     c = 2
```

scanf Format Codes

Code	Format
%c	character
%s	character string
%d	decimal integers
%i	decimal integers
%f	floating pointer number (printf)
%lf	floating pointer number (scanf)
%e	scientific notation (with lower case e)
%E	scientific notation (with upper case E)
%g	the shorter of %f and %e
%G	the shorter of %f and %E
%u	unsigned decimal integer
%o	unsigned octal number
%x	unsigned hexadecimal number (with lower case letters)
%X	unsigned hexadecimal number (with upper case letters)
%p	pointer
%%	print %

Formatted Output

- **printf()** function is used to print output on the console screen.
- Syntax:

  printf(“control string”, arg1,arg2, … , argn);

- Similar to scanf() function, format specifier is used inside control string to tell compiler about variables being printed.

Output of Integer Numbers

- Syntax:

  printf(“ %nd ”, var_name);

- Here, n specifies minimum field width for output.

Format	***Output***
printf(“ %d ”,1234);	1 2 3 4
printf(“ %7d ”,1234);	_ _ _ 1 2 3 4
printf(“ %2d ”,1234);	1 2 3 4
printf(“ %-7d ”,1234);	1 2 3 4 _ _ _
printf(“ %07d ”,1234);	0 0 0 1 2 3 4

Output of Real Numbers

- Syntax:

  printf(“ %{n.m}f ”, var_name); OR

  printf(“ %{w.p}e ”, var_name);

- Here, **n** specifies minimum field width for output and **m** specifies decimal digits after floating point notation.
- We can also decide width and precision at runtime using variables.
- **printf(“%*.*f”, width, precision, number);**

Output of Real Numbers

Format	*Output*
printf("%7.4f",y)	98.7654
printf("%7.2f",y)	98.77
printf("%-7.2f",y)	98.77
printf("%f",y)	98.7654
printf("%10.2e",y)	9.88e+01
printf("%11.4e",-y)	-9.8765e+01
printf("%-10.2e",y)	9.88e+01
printf("%e",y)	9.876540e+01

Output Format Codes & Flags

specifier	argument
d or i	Singed INT
u	Unsinged INT
o	Unsigned Octal
x or X	Unsigned Hexadecimal (lower/upper case)
e or E	Floating Point
g or G	Shortest Representation
a or A	Hexadecimal Floating Point (lower/upper case)
c	Character
s	String of characters
p	Pointer address

flag	description
-	Left-justify (default: right)
+	Force print + sign with positives
(space)	Add space, if no sign before value
#	Precede o, x, or X with 0
0	Left pad number with zeros

String I/O

- String is nothing but the collection of characters.
- "**gets()**" and "**puts()**" functions are used to take user input in string form.
- Syntax:

  gets(string_var) **puts(string_var);**

- Alternative Representation:

  scanf("%s",string_var); **printf("%s",string_var);**

Printing of Single Character

- Use the following format to print the single character using following format:

  %wc

- The character will be **right justified** in **width of w** columns.

Common Output Flags

Flag	*Meaning*
–	Output is left-justified within the field. Remaining field will be blank.
+	+ or – will precede the signed numeric item.
0	Causes leading zeros to appear.
# (with o or x)	Causes octal and hex items to be preceded by O and Ox, respectively.
# (with e, f or g)	Causes a decimal point to be present in all floating point numbers, even if it is whole number. Also prevents the truncation of trailing zeros in g-type conversion.

Enhancing Readability of Output

- Correctness and clarity in output is very important.
- Some common measures are as follow:
  - Provide enough blank space between two numbers.
  - Introduce appropriate headings and variable names in output.
  - Print special extra messages whenever a peculiar condition occurs in the output.
  - Introduce extra blank lines between important sections of output.

Thank You